

Exercise 19.1

1. Evaluate the following

a) $\int x^3 dx$

$$= \frac{x^{3+1}}{3+1} + C$$

$$= \frac{x^4}{4} + C$$

b) $\int 2x^{-3} dx$

$$= 2 \int x^{-3} dx$$

$$= 2 \frac{x^{-3+1}}{-3+1} + C$$

$$= \cancel{2} \frac{x^{-2}}{\cancel{-2}} + C$$

$$= -x^{-2} + C$$

$$= \frac{-1}{x^2} + C$$

c) $\int (x^2 + 2) dx$

$$= \int x^2 dx + \int 2 dx$$

$$= \frac{x^{2+1}}{2+1} + 2x + C$$

$$= \frac{x^3}{3} + 2x + C$$

d) $\int (x+1)(x-4) dx$

$$= \int (x^2 - 4x + x - 4) dx$$

$$= \int (x^2 - 3x - 4) dx$$

$$= \int x^2 dx - \int 3x dx + \int 4 dx$$

$$= \frac{x^{2+1}}{2+1} - 3 \frac{x^{1+1}}{1+1} + 4x + C$$

$$= \frac{x^3}{3} - 3 \frac{x^2}{2} + 4x + C$$

$$= \frac{x^3}{3} - \frac{3}{2} x^2 + 4x + C$$

$$e) \int \sqrt{x} (x^2 - 5) dx$$

$$= \int x^{1/2} (x^2 - 5) dx$$

$$= \int (x^{5/2} - 5x^{1/2}) dx$$

$$= \int x^{5/2} dx - \int 5x^{1/2} dx$$

$$= \frac{x^{5/2+1}}{5/2+1} - 5 \frac{x^{1/2+1}}{1/2+1} + C$$

$$= \frac{x^{7/2}}{7/2} - 5 \frac{x^{3/2}}{3/2} + C$$

$$= \frac{2}{7} x^{7/2} - \frac{10}{3} x^{3/2} + C$$

$$f) \int \frac{ax^2 + bx + c}{x^2} dx$$

$$= \int \left(a + \frac{b}{x} + \frac{c}{x^2} \right) dx$$

$$= \int (a + bx^{-1} + cx^{-2}) dx$$

$$= \int a dx + \int bx^{-1} dx + \int cx^{-2} dx$$

$$= ax + b \ln x + c \frac{x^{-2+1}}{-2+1} + K$$

$$= an + b \ln x + \frac{c x^{-1}}{-1} + K$$

$$= an + b \ln x - \frac{c}{x} + K$$

$$g) \int \left(x^2 + \frac{1}{x^2} \right)^2 dx$$

$$= \int \left(x^4 + 2 \cdot x^2 \cdot \frac{1}{x^2} + \frac{1}{x^4} \right) dx$$

$$= \int (x^4 + 2 + x^{-4}) dx$$

$$= \int x^4 dx + \int 2 dx + \int x^{-4} dx$$

$$= \frac{x^5}{5} + 2x + \frac{x^{-4+1}}{-4+1} + C$$

$$= \frac{x^5}{5} + 2x + \frac{x^{-3}}{-3} + C$$

$$= \frac{x^5}{5} + 2x - \frac{1}{3x^3} + C$$

2.

2. Evaluate

$$a) \int (3x+5)^4 dx$$

$$= \frac{(3x+5)^{4+1}}{3(4+1)} + C$$

$$= \frac{(3x+5)^5}{15} + C$$

2.

$$b) \int (5-6x)^7 dx$$

$$= \frac{(5-6x)^{7+1}}{-6(7+1)} + C$$

$$= \frac{(5-6x)^8}{-48} + C$$

$$c) \int \sqrt{2x-7} dx$$

$$= \int (2x-7)^{1/2} dx$$

$$= \frac{(2x-7)^{1/2+1}}{2\left(\frac{1}{2}+1\right)} + C$$

$$= \frac{(2x-7)^{3/2}}{3} + C$$

$$d) \int \frac{L}{\sqrt{2x+1}} dx$$

$$= \int \frac{1}{(2x+1)^{1/2}} dx$$

$$= \int (2x+1)^{-1/2} dx$$

$$= \frac{(2x+1)^{-1/2+1}}{2\left(-\frac{1}{2}+1\right)} + C$$

$$= \frac{(2x+1)^{1/2}}{1} + C$$

$$= (2x+1)^{1/2} + C$$

$$\begin{aligned}
 e) & \int \left[2 + \frac{1}{(x-1)^2} \right] dx \\
 &= \int 2 dx + \int (x-1)^{-2} dx \\
 &= 2x + \frac{(x-1)^{-2+1}}{-2+1} + C \\
 &= 2x + \frac{(x-1)^{-1}}{-1} + C \\
 &= 2x - (x-1)^{-1} + C \\
 &= 2x - \frac{1}{x-1} + C
 \end{aligned}$$

$$\begin{aligned}
 f) & \int \frac{x-2}{x^2+x-6} dx \\
 &= \int \frac{x-2}{x^2+3x-2x-6} dx \\
 &= \int \frac{x-2}{(x+3)(x-2)} dx \\
 &= \int \frac{\cancel{x-2}}{(x+3)\cancel{(x-2)}} dx = \int \frac{1}{x+3} dx \\
 &= \ln(x+3) + C
 \end{aligned}$$

$$g) \int \frac{x+3}{x-3} dx$$

$$= \int \frac{(x-3)+6}{x-3} dx$$

$$= \int 1 + \frac{6}{x-3} dx$$

$$= \int 1 dx + \int \frac{6}{x-3} dx$$

$$= x + 6 \int \frac{1}{x-3} dx$$

$$= x + 6 \ln(x-3) + C$$

$$h) \int \frac{x^2+5}{x+2} dx$$

$$= \int \frac{x^2+4+9}{x+2} dx$$

$$= \int \frac{x^2-4}{x+2} + \frac{9}{x+2} dx$$

$$= \int \frac{(x+2)(x-2)}{(x+2)} + \frac{9}{x+2} dx$$

$$= \frac{(x-2)^2}{2} + 9 \ln(x+2) + C$$

$$= \frac{x^2}{2} - \frac{4x}{2} + \frac{4}{2} + 9 \ln(x+2) + C$$

$$= \frac{x^2}{2} - 2x + 2 + 9 \ln(x+2) + C$$

$$i) \int \frac{x^2 + 3x + 3}{x+1} dx$$

$$= \int \frac{x^2 + 2x + x + 2 + 1}{x+1} dx$$

$$= \int \frac{x(x+2) + 1(x+2)}{x+1} + \frac{1}{x+1} dx$$

$$= \int x+2 + \frac{1}{x+1} dx$$

$$= \frac{(x+2)^2}{2} + \ln(x+1) + C$$

$$= \frac{x^2}{2} + \frac{4x}{2} + \frac{4}{2} + \ln(x+1) + C$$

$$= \frac{x^2}{2} + 2x + 2 + \ln(x+1) + C$$

3. Evaluate

~~$$\int \frac{1}{\sqrt{x+a} + \sqrt{x+b}} dx$$~~

$$a) \int \frac{1}{\sqrt{x+a} + \sqrt{x+b}} dx$$

$$= \int \frac{1}{\sqrt{x+a} + \sqrt{x+b}} \times \frac{\sqrt{x+a} - \sqrt{x+b}}{\sqrt{x+a} - \sqrt{x+b}} dx$$

$$= \int \frac{\sqrt{x+a} - \sqrt{x+b}}{x+a-x-b} dx$$

$$= \frac{1}{a-b} \int (x+a)^{1/2} - (x+b)^{1/2} dx$$

$$= \frac{1}{a-b} \left[\int (x+a)^{1/2+1} dx - \int (x+b)^{1/2+1} dx \right]$$

$$= \frac{1}{a-b} \left[\frac{(x+a)^{1/2+1}}{1/2+1} - \frac{(x+b)^{1/2+1}}{1/2+1} \right] + C$$

$$= \frac{1}{a-b} \left[\frac{2(x+a)^{3/2}}{3} - \frac{2(x+b)^{3/2}}{3} \right] + C$$

$$= \frac{2(x+a)^{3/2} - 2(x+b)^{3/2}}{3(a-b)} + C$$

b) $\int \frac{1}{\sqrt{x+a} + \sqrt{x}} dx$

$$= \int \left[\frac{1}{\sqrt{x+a} + \sqrt{x}} \times \frac{\sqrt{x+a} - \sqrt{x}}{\sqrt{x+a} - \sqrt{x}} \right] dx$$

$$= \int \left[\frac{\sqrt{x+a} - \sqrt{x}}{x+a-x} \right] dx$$

$$= \frac{1}{a} \int [(x+a)^{1/2} - x^{1/2}] dx$$

$$= \frac{1}{a} \left[\frac{(x+a)^{1/2+1}}{\frac{1}{2}+1} - \frac{x^{1/2+1}}{\frac{1}{2}+1} \right] + C$$

$$= \frac{1}{a} \left[\frac{(x+a)^{3/2}}{3/2} - \frac{x^{3/2}}{3/2} \right] + C$$

$$= \frac{2}{3a} [(x+a)^{3/2} - x^{3/2}] + C$$

c) $\int \frac{1}{\sqrt{3x+a} + \sqrt{3x-b}} dx$

$$= \int \left[\frac{1}{\sqrt{3x+a} + \sqrt{3x-b}} \times \frac{\sqrt{3x+a} - \sqrt{3x-b}}{\sqrt{3x+a} - \sqrt{3x-b}} \right] dx$$

$$= \int \left[\frac{\sqrt{3x+a} - \sqrt{3x-b}}{3x+a-3x+b} \right] dx$$

$$= \frac{1}{a+b} \left[\int (3x+a)^{1/2} dx - \int (3x-b)^{1/2} dx \right]$$

$$= \frac{1}{a+b} \left[\frac{(3x+a)^{3/2}}{(3/2) \times 3} - \frac{(3x-b)^{3/2}}{(3/2) \times 3} \right] + c$$

$$= \frac{2}{9(a+b)} \left[(3x+a)^{3/2} - (3x-b)^{3/2} \right] + c$$

$$d) \int \frac{1}{\sqrt{2x+1} + \sqrt{2x-3}} dx$$

$$= \int \left[\frac{1}{\sqrt{2x+1} + \sqrt{2x-3}} \times \frac{\sqrt{2x+1} - \sqrt{2x-3}}{\sqrt{2x+1} - \sqrt{2x-3}} \right] dx$$

$$= \int \left[\frac{\sqrt{2x+1} - \sqrt{2x-3}}{2x+1 - 2x+3} \right] dx$$

$$= \int \left[\frac{(2x+1)^{1/2} - (2x-3)^{1/2}}{4} \right] dx$$

$$= \frac{1}{4} \left[\int (2x+1)^{1/2} dx - \int (2x-3)^{1/2} dx \right]$$

$$= \frac{1}{4} \left[\frac{(2x+1)^{3/2}}{\frac{3}{2} \times 2} - \frac{(2x-3)^{3/2}}{\frac{3}{2} \times 2} \right] + c$$

$$= \frac{1}{12} \left[(2x+1)^{3/2} - (2x-3)^{3/2} \right] + c$$

5. Evaluate.

$$a) \int \sin 4x dx$$

$$= \frac{-\cos 4x}{4} + c$$

$$\begin{aligned} b) \int \sin(4x-1) dx &= -\frac{\cos(4x-1)}{(4x-1)'} + C \\ &= -\frac{1}{4} \cos(4x-1) + C \end{aligned}$$

$$\begin{aligned} c) \int \sin a^2 x dx &= \frac{-\cos a^2 x}{a^2} + C \end{aligned}$$

$$\begin{aligned} d) \int \sin^2 ax dx &= \int \left[\frac{1 - \cos 2ax}{2} \right] dx \\ &= \frac{1}{2} \left[\int 1 dx - \int \cos 2ax dx \right] \\ &= \frac{1}{2} \left[x - \frac{\sin 2ax}{2a} \right] + C \\ &= \frac{1}{2} x - \frac{1}{4a} \sin 2ax + C \end{aligned}$$

$$\begin{aligned} e) \int \cos^2 ax dx &= \int \left[\frac{1 + \cos 2ax}{2} \right] dx \\ &= \frac{1}{2} \int (1 + \cos 2ax) dx \\ &= \frac{1}{2} \left[\int 1 dx + \int \cos 2ax dx \right] \\ &= \frac{1}{2} \left[x + \frac{\sin 2ax}{2a} \right] + C \\ &= \frac{x}{2} + \frac{1}{4a} \sin 2ax + C \end{aligned}$$

$$\begin{aligned}
 f) \int \sin^4 ax \, dx &= \int \frac{1}{4} \times 4 \sin^4 ax \, dx \\
 &= \frac{1}{4} \left[\int (2 \sin^2 ax)^2 \, dx \right] \\
 &= \frac{1}{4} \left[\int (1 - \cos 2ax)^2 \, dx \right] \\
 &= \frac{1}{4} \int [1 - 2 \cos 2ax + \cos^2 2ax] \, dx \\
 &= \frac{1}{4} \left[\int 1 \, dx - \int 2 \cos 2ax \, dx + \int \cos^2 2ax \, dx \right] \\
 &= \frac{1}{4} \left[x - \frac{2 \sin 2ax}{2a} + \int \frac{(1 + \cos 4ax)}{2} \, dx \right] \\
 &= \frac{x}{4} - \frac{\sin 2ax}{4a} + \frac{1}{4} \left[\int \frac{1}{2} \, dx + \frac{1}{2} \int \cos 4ax \, dx \right] \\
 &= \frac{x}{4} - \frac{\sin 2ax}{4a} + \frac{1}{4} \left[\frac{x}{2} + \frac{\sin 4ax}{2 \times 4a} \right] + C \\
 &= \frac{x}{4} - \frac{\sin 2ax}{4a} + \frac{x}{8} + \frac{\sin 4ax}{32a} + C \\
 &= \frac{3x}{8} - \frac{\sin 2ax}{4a} + \frac{\sin 4ax}{32a} + C
 \end{aligned}$$

$$\begin{aligned}
 g) \int \cos^4 ax \, dx &= \int \left[\frac{1}{4} \times 4 \cos^4 ax \right] \, dx \\
 &= \int \left[\frac{1}{4} \times (2 \cos^2 ax)^2 \right] \, dx \\
 &= \frac{1}{4} \int (\cos 2ax + 1)^2 \, dx
 \end{aligned}$$

$$= \frac{1}{4} \int (1 + 2 \cos 2an + \cos^2 2an) dn$$

$$= \frac{1}{4} \left[\int 1 dn + \int 2 \cos 2an dn + \int \cos^2 2an dn \right]$$

$$= \frac{1}{4} \left[n + \frac{2 \sin 2an}{2a} + \int \frac{\cos 4an + 1}{2} dn \right]$$

$$= \frac{1}{4} \left[n + \frac{\sin 2an}{a} + \frac{1}{2} \left(\int \cos 4an dn + \int 1 dn \right) \right]$$

$$= \frac{1}{4} \left[n + \frac{\sin 2an}{a} + \frac{1}{2} \left(\frac{\sin 4an}{4a} + n \right) \right] + C$$

$$= \frac{1}{4} \left[n + \frac{\sin 2an}{a} + \frac{1}{8a} \sin 4an + \frac{n}{2} \right] + C$$

$$= \frac{n}{4} + \frac{\sin 2an}{4a} + \frac{1}{32a} \sin 4an + \frac{n}{8} + C$$

$$= \frac{3n}{8} + \frac{\sin 2an}{4a} + \frac{1}{32a} \sin 4an + C$$

$$b) \int \sin^3 bx dx$$

$$= \frac{1}{4} \int (3 \sin bx - \sin 3bx) dx$$

$$= \frac{1}{4} \left[\int 3 \sin bx dx - \int \sin 3bx dx \right]$$

$$= \frac{1}{4} \left[\frac{3(-\cos bx)}{b} - \frac{(-\cos 3bx)}{3b} \right] + C$$

$$= \frac{1}{4} \left[\frac{-3 \cos bx}{b} + \frac{\cos 3bx}{3b} \right] + C$$

$$= \frac{1}{4b} \left[\frac{\cos 3bx}{3} - 3 \cos bx \right] + C$$

$$= \frac{1}{12b} [\cos 3bx - 9 \cos bx] + C$$

$$i) \int \frac{1}{\sin^2 x} dx$$

$$= \int \operatorname{cosec}^2 x dx$$

$$= -\cot x + C$$

$$j) \int \frac{1}{\sin^2 x \cdot \cos^2 x} dx$$

$$= \int \frac{\sin^2 x + \cos^2 x}{\sin^2 x \cdot \cos^2 x} dx$$

$$= \int \left(\frac{\sin^2 x}{\sin^2 x \cdot \cos^2 x} + \frac{\cos^2 x}{\sin^2 x \cdot \cos^2 x} \right) dx$$

$$= \int \left(\frac{1}{\cos^2 x} + \frac{1}{\sin^2 x} \right) dx$$

$$= \int (\sec^2 x + \operatorname{cosec}^2 x) dx$$

$$= \int \sec^2 x dx + \int \operatorname{cosec}^2 x dx$$

$$= \tan x - \cot x + C$$

$$k) \int \frac{1}{\sec^2 x \cdot \tan^2 x} dx$$

$$= \int \frac{\sec^2 x - \tan^2 x}{\sec^2 x \cdot \tan^2 x} dx$$

$$= \int \left(\frac{1}{\tan^2 x} - \frac{1}{\sec^2 x} \right) dx$$

$$= \int \left(\frac{\cos^2 x}{\sin^2 x} - \cos^2 x \right) dx$$

$$= \int (\cot^2 x - \cos^2 x) dx$$

$$= \int (\operatorname{cosec}^2 x - 1) dx - \int \cos^2 x dx$$

$$= \int \operatorname{cosec}^2 x dx - \int 1 dx - \frac{1}{2} \int (\cos 2x + 1) dx$$

$$= -\cot x - x - \frac{1}{2} \left[\int \cos 2x dx + \int 1 dx \right]$$

$$= -\cot x - x - \frac{1}{2} \left[\frac{\sin 2x}{2} + x \right] + C$$

$$= -\cot x - x - \frac{1}{4} \sin 2x - \frac{1}{2} x + C$$

$$= -\cot x - \frac{3}{2} x - \frac{1}{4} \sin 2x + C$$

$$L) \int \cot^2(ax+b) dx$$

$$= \int (\operatorname{cosec}^2(ax+b) - 1) dx$$

$$= \int \operatorname{cosec}^2(ax+b) dx - \int 1 dx$$

$$= \frac{-\cot(ax+b)}{a} - x + C$$

6) Evaluate

$$a) \int \sqrt{1 - \cos x} dx$$

$$= \int \sqrt{2 \sin^2 x/2} dx$$

$$= \int (\sqrt{2} \sin x/2) dx$$

$$= \sqrt{2} \int \sin x/2 dx$$

$$= \frac{\sqrt{2}(-\cos x/2)}{1/2} + C$$

$$= -2\sqrt{2} \cos x/2 + C$$

$$b) \int \sqrt{1 + \sin 2x} dx$$

$$= \int \sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cos x} dx$$

$$= \int \sqrt{(\sin x + \cos x)^2} dx$$

$$= \int (\sin x + \cos x) dx$$

$$= \int \sin x dx + \int \cos x dx$$

$$= -\cos x + \sin x + C$$

$$c) \int \sqrt{1 - \sin(ax+b)} \, dx$$

$$= \int \sqrt{\sin^2\left(\frac{ax+b}{2}\right) + \cos^2\left(\frac{ax+b}{2}\right) - 2\sin\left(\frac{ax+b}{2}\right) \cdot \cos\left(\frac{ax+b}{2}\right)} \, dx$$

$$= \int \sqrt{\left[\sin\left(\frac{ax+b}{2}\right) - \cos\left(\frac{ax+b}{2}\right)\right]^2} \, dx$$

$$= \int \left[\sin\left(\frac{ax+b}{2}\right) - \cos\left(\frac{ax+b}{2}\right)\right] \, dx$$

$$= \int \sin\left(\frac{ax+b}{2}\right) \, dx - \int \cos\left(\frac{ax+b}{2}\right) \, dx$$

$$= \underbrace{-\cos\left(\frac{ax+b}{2}\right)}_{a/2} - \underbrace{\sin\left(\frac{ax+b}{2}\right)}_{a/2} + C$$

$$= -\frac{2}{a} \left[\sin\left(\frac{ax+b}{2}\right) + \cos\left(\frac{ax+b}{2}\right) \right] + C$$

$$d) \int \frac{\cos 2x}{\cos x - \sin x} \, dx$$

$$= \int \frac{\cos^2 x - \sin^2 x}{\cos x - \sin x} \, dx$$

$$= \int \frac{(\cos x + \sin x)(\cos x - \sin x)}{(\cos x - \sin x)} \, dx$$

$$= \int \cos x \, dx + \int \sin x \, dx$$

$$= +\sin x - \cos x + C$$

$$= \sin x - \cos x + C$$

$$e) \int \frac{1}{1 - \cos nx} dx$$

$$\cos b (d + \cos) \sin^2 - 1 \dots$$

$$\left(\frac{d + \cos}{s} \right) = \int \frac{1}{2 \sin^2 \frac{nx}{2}} dx = \left(\frac{d + \cos}{s} \right)^{\cos} + \left(\frac{d + \cos}{s} \right)^{\sin^2} \dots$$

$$= \int \frac{\operatorname{cosec}^2 \frac{nx}{2}}{2} dx$$

$$= \frac{1}{2} \int \operatorname{cosec}^2 \frac{nx}{2} dx$$

$$= \frac{1}{2} \left(-\cot \frac{nx}{2} \right) + C$$

$$= -\frac{1}{2} \times \frac{2}{n} \cot \frac{nx}{2} + C$$

$$= -\frac{1}{n} \cot \frac{nx}{2} + C$$

$$f) \int \frac{1}{1 - \sin ax} dx$$

$$= \int \left[\frac{1}{1 - \sin ax} \times \frac{1 + \sin ax}{1 + \sin ax} \right] dx$$

$$= \int \left[\frac{1 + \sin ax}{1 - \sin^2 ax} \right] dx$$

$$= \int \left[\frac{1 + \sin ax}{\cos^2 ax} \right] dx$$

$$= \int \left[\frac{1}{\cos^2 ax} + \frac{\sin ax}{\cos ax} \times \frac{1}{\cos ax} \right] dx$$

$$= \int [\sec^2 ax + \tan ax \times \sec ax] dx$$

$$= \int \sec^2 ax dx + \int \tan ax \cdot \sec ax dx$$

$$= \frac{\tan a x}{a} + \frac{\sec a x}{a} + C$$

$$= \frac{1}{a} (\tan a x + \sec a x) + C$$

⑦ Evaluate

a) $\int \sin 4x \cdot \sin 2x \, dx$

$$= \frac{1}{2} \int 2 \sin 4x \cdot \sin 2x \, dx$$

$$= \frac{1}{2} \int [\cos(4x-2x) - \cos(4x+2x)] \, dx$$

$$= \frac{1}{2} \int [\cos 2x - \cos 6x] \, dx$$

$$= \frac{1}{2} \left[\int \cos 2x \, dx - \int \cos 6x \, dx \right]$$

$$= \frac{1}{2} \left[\frac{\sin 2x}{2} - \left(\frac{\sin 6x}{6} \right) \right]$$

$$= \frac{1}{4} \sin 2x - \frac{1}{12} \sin 6x$$

b) $\int \sin 4x \cdot \cos 2x \, dx$

$$= \frac{1}{2} \int 2 \sin 4x \cdot \cos 2x \, dx$$

$$= \frac{1}{2} \int [\sin(4x+2x) + \sin(4x-2x)] \, dx$$

$$= \frac{1}{2} \int (\sin 6x + \sin 2x) \, dx$$

$$= \frac{1}{2} \left[\int \sin 6x \, dx + \int \sin 2x \, dx \right]$$

$$= \frac{1}{2} \left[-\frac{\cos 6x}{6} - \frac{\cos 2x}{2} \right] + C$$

$$= -\frac{\cos 6x}{12} - \frac{\cos 2x}{4} + C$$

c) $\int \cos 4x \cdot \cos 2x \, dx$

$$= \frac{1}{2} \int 2 \cos 4x \cdot \cos 2x \, dx$$

$$= \frac{1}{2} \int [\cos(4x+2x) + \cos(4x-2x)] \, dx$$

$$= \frac{1}{2} \int [\cos 6x + \cos 2x] \, dx$$

$$= \frac{1}{2} \left[\int \cos 6x \, dx + \int \cos 2x \, dx \right]$$

$$= \frac{1}{2} \left[\frac{\sin 6x}{6} + \frac{\sin 2x}{2} \right] + C$$

$$= \frac{1}{12} \sin 6x + \frac{1}{4} \sin 2x + C$$

d) $\int \sin 6x \cdot \cos 8x \, dx$

$$= \frac{1}{2} \int 2 \sin 6x \cdot \cos 8x \, dx$$

$$= \frac{1}{2} \int 2 \cos 8x \cdot \sin 6x \, dx$$

$$= \frac{1}{2} \int [\sin(8x+6x) - \sin(8x-6x)] \, dx$$

$$= \frac{1}{2} \int (\sin 14x - \sin 2x) \, dx$$

$$= \frac{1}{2} \left[\int \sin 14x \, dx - \int \sin 2x \, dx \right]$$

$$= \frac{1}{2} \left[\frac{(-\cos 14x)}{14} - \frac{(-\cos 2x)}{2} \right] + C$$

$$= -\frac{1}{28} \cos 14x + \frac{1}{4} \cos 2x + C$$

$$= \frac{1}{4} \cos 2x - \frac{1}{28} \cos 14x + C$$

8.

a) $\int (e^{ax} + e^{-bx}) dx$

$$= \int e^{ax} dx + \int e^{-bx} dx$$

$$= \frac{e^{ax}}{a} + \frac{e^{-bx}}{-b} + C$$

$$= \frac{e^{ax}}{a} - \frac{e^{-bx}}{b} + C$$

b) $\int (e^{ax} + e^{-ax})^2 dx$

$$= \int (e^{2ax} + 2 \cdot e^{ax} \cdot e^{-ax} + e^{-2ax}) dx$$

$$= \int (e^{2ax} + 2 + e^{-2ax}) dx$$

$$= \int e^{2ax} dx + \int 2 dx + \int e^{-2ax} dx$$

$$= \frac{e^{2ax}}{2a} + x + \frac{e^{-2ax}}{-2a} + C$$

$$= \frac{e^{2ax}}{2a} + x - \frac{e^{-2ax}}{2a} + C$$

c) $\int \frac{e^{3x} + e^x - 1}{e^x} dx$

$$= \int \left[\frac{e^{3x}}{e^x} + \frac{e^x}{e^x} - \frac{1}{e^x} \right] dx$$

$$= \int [e^{2x} + 1 - e^{-x}] dx$$

$$= \int e^{2x} dx + \int 1 dx - \int e^{-x} dx$$

$$= \frac{e^{2x}}{2} + x - \frac{e^{-x}}{-1} + C$$

$$= \frac{e^{2x}}{2} + x + \frac{e^{-x}}{1} + C$$

$$= \frac{e^{2x}}{2} + e^{-x} + x + C$$

✓

9) Evaluate the following definite integral

$$a) \int_2^3 x^2 dx$$

$$= \left[\int x^2 dx \right]_2^3$$

$$= \left[\frac{x^3}{3} \right]_2^3$$

$$= \frac{3^3}{3} - \frac{2^3}{3}$$

$$= \frac{27}{3} - \frac{8}{3}$$

$$= 9 - \frac{8}{3}$$

$$= \frac{27-8}{3}$$

$$= \frac{19}{3}$$

✓

$$b) \int_{-1}^3 (3x^2 + 2x + 7) dx$$

$$= \left[\int (3x^2 + 2x + 7) dx \right]_{-1}^3$$

$$= \left[\int 3x^2 dx + \int 2x dx + \int 7 dx \right]_{-1}^3$$

$$= \left[\frac{3x^3}{3} + \frac{2x^2}{2} + 7x \right]_{-1}^3$$

$$= [x^3 + x^2 + 7x]_{-1}^3$$

$$= 3^3 + 3^2 + 7 \times 3 - (-1)^3 - (-1)^2 - 7 \times -1$$

$$= 27 + 9 + 21 + 1 - 1 + 7$$

$$= 64$$

Ans.

$$c) \int_0^1 (2x+1)^3 dx$$

$$= \left[\int (2x+1)^3 dx \right]_0^1$$

$$= \left[\frac{(2x+1)^4}{2 \times 4} \right]_0^1$$

$$= \left[\frac{(2x+1)^4}{8} \right]_0^1$$

$$= \frac{(2 \times 1 + 1)^4}{8} - \frac{(2 \times 0 + 1)^4}{8}$$

$$= \frac{3^4}{8} - \frac{1}{8}$$

$$= \frac{81-1}{8} = \frac{80}{8}$$

$$= 10$$

Ans.

$$\begin{aligned}
 & \int_1^2 \frac{1}{2x-1} dx \\
 &= \int_1^2 \frac{1}{2(x-\frac{1}{2})} dx \\
 &= \frac{1}{2} \int_1^2 \frac{1}{x-\frac{1}{2}} dx \\
 &= \frac{1}{2} [\ln(x-\frac{1}{2})]_1^2 \\
 &= \frac{1}{2} (\ln(2-\frac{1}{2}) - \ln(1-\frac{1}{2})) \\
 &= \frac{1}{2} (\ln(\frac{3}{2}) - \ln(\frac{1}{2})) \\
 &= \frac{1}{2} (\ln 3 - \ln 1) \\
 &= \frac{1}{2} \ln 3
 \end{aligned}$$

$$\begin{aligned}
 & \int_1^2 \frac{1}{2x-1} dx \\
 &= \frac{1}{2} [\ln(2x-1)]_1^2 \\
 &= \frac{1}{2} (\ln(2 \times 2 - 1) - \ln(2 \times 1 - 1)) \\
 &= \frac{1}{2} (\ln 3 - \ln 1) \\
 &= \frac{1}{2} \ln 3 - 0 \quad \because \ln 1 = 0 \\
 &= \frac{1}{2} \ln 3 \\
 &= \ln 3^{1/2} \\
 &= \ln \sqrt{3}
 \end{aligned}$$

$$\begin{aligned}
 & \int_0^{\pi/3} \sec^2 x dx \\
 &= [\tan x]_0^{\pi/3} \\
 &= \tan \frac{\pi}{3} - \tan 0 \\
 &= \sqrt{3} - 0 \\
 &= \sqrt{3}
 \end{aligned}$$

$$= \sqrt{3} - 0$$

$$= \sqrt{3}$$

$$f) \int_{\pi/4}^{\pi/2} \cot^2 x \, dx$$

$$= \int_{\pi/4}^{\pi/2} (\operatorname{cosec}^2 x - 1) \, dx$$

$$= \int_{\pi/4}^{\pi/2} [\operatorname{cosec}^2 x \, dx - 1 \, dx]$$

$$= \left[\int \operatorname{cosec}^2 x \, dx - \int 1 \, dx \right]_{\pi/4}^{\pi/2}$$

$$= \left[-\cot x - x \right]_{\pi/4}^{\pi/2}$$

$$= \left(-\cot \frac{\pi}{2} - \frac{\pi}{2} \right) - \left(-\cot \frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= 0 - \frac{\pi}{2} + \frac{\pi}{4} + 1$$

$$= 1 - \frac{\pi}{4}$$

$$g) \int_0^{\pi/2} \frac{1}{1 + \sin x} \, dx$$

$$= \int_0^{\pi/2} \left(\frac{1}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x} \right) dx$$

$$= \int_0^{\pi/2} \left(\frac{1 - \sin x}{\cos^2 x} \right) dx$$

$$g) \int_0^{\pi/2} \frac{1}{1 + \sin x} dx$$

$$= \int_0^{\pi/2} \frac{1}{1 + \cos\left(\frac{\pi}{2} - x\right)} dx$$

$$= \int_0^{\pi/2} \frac{1}{2 \cos^2\left(\frac{\pi}{2} - x\right)} dx$$

$$= \frac{1}{2} \int_0^{\pi/2} \sec^2\left(\frac{\pi}{2} - x\right) dx$$

$$= \frac{1}{2} \left[\tan\left(\frac{\pi}{2} - x\right) \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\tan\left(\frac{\pi}{2} - \frac{\pi}{2}\right) - \tan\left(\frac{\pi}{2} - 0\right) \right]$$

$$= \frac{1}{2} \times -1 \left[\tan\left(\frac{\pi}{2} - \frac{\pi}{2}\right) - \tan\left(\frac{\pi}{2} - 0\right) \right]$$

$$= -1 (0 - 1)$$

$$= -1 \times -1$$

$$= 1$$

Ans.

$$b) \int_0^{\pi/3} \frac{1}{\cos x + 1} dx$$

$$= \int_0^{\pi/3} \frac{1}{2\cos^2 x/2} dx$$

$$= \int_0^{\pi/3} \frac{\sec^2 x/2}{2} dx$$

$$= \frac{1}{2} \left[\int \sec^2 x/2 dx \right]_0^{\pi/3}$$

$$= \frac{1}{2} \left[\tan \frac{x}{2} \right]_0^{\pi/3}$$

$$= \frac{1}{2} \times \frac{2}{1} \left[\tan \frac{\pi}{6} - \tan 0 \right]$$

$$= \frac{1}{\sqrt{3}} - 0$$

$$= \frac{1}{\sqrt{3}}$$